4.6 Karyotyping

Karyotyping is a technique through which a complete set of chromosomes is separated from a cell and the chromosomes are arranged in pairs. An idiogram refers to a diagrammatic representation of chromosomes.

Preparation of Karyotype

Tjio and Levan (1960) described a simple method of culturing lymphocytes from the human blood. Mitosis is induced followed by addition of colchicine to arrest cell division at metaphase stage and the suitable spread of metaphase chromosomes is photographed. The individual chromosomes are cut from the photograph and are arranged in an orderly fashion in homologous pairs. This arrangement is called a **karyotype**. Chromosome banding permits structural definitions and differentiation of chromosomes.

Applications of Karyotyping

- It helps in gender identification.
- It is used to detect the chromosomal aberrations like deletion, duplication, translocation, nondisjunction of chromosomes.
- It helps to identify the abnormalities of chromosomes like aneuploidy.
- It is also used in predicting the evolutionary relationships between species.
- Genetic diseases in human beings can be detected by this technique.

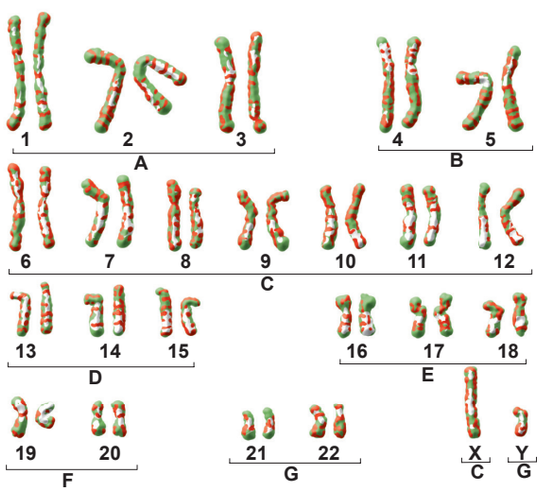


Fig. 4.5 Human karyotype (male)

Human Karyotype

Depending upon the position of the centromere and relative length of two arms, human chromosomes are of three types: Metacentric, sub metacentric and acrocentric. The photograph of chromosomes are arranged in the order of descending length in groups from A to G **(Fig. 4.5).**

4.7 Pedigree Analysis

Pedigree is a "family tree", drawn with standard genetic symbols, showing the inheritance pathway for specific phenotypic characters.**(Fig. 4.6).** Pedigree analysis is the study of traits as they have appeared in a given family line for several past generations.

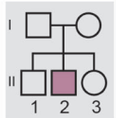
Symbol	Explanation	Symbol	Explanation
□	Male	■ ♂ ● ♀	Affected individuals
○	Female	◧ ♂ ◐ ♀	Heterozygotes for autosomal recessives
□—○	Mating	◉ ♀	Carrier of sex-linked recessives
	Parents and children (1 boy: 1 girl in order of birth)	⊘	Death
	Dizygotic twins	|	Abortion or still birth (sex unspecified)
	Monozygotic twins		Propositus (proband)
◇	Sex unspecified		Method of identifying persons in a pedigree : here the propositus in child 2 in generation 2 or II 2
2 ③	Number of children of sex indicated	□—○	Consanguineous marriage

Fig. 4.6 Symbols commonly used in pedigree charts

Genetic Disorders

A genetic disorder is a disease or syndrome that is caused by an abnormality in an individual DNA. Abnormalities can range from a small mutation in a single gene to the addition or subtraction of an entire chromosome or even a set of chromosomes. Genetic disorders are of two types namely, Mendelian disorders and chromosomal disorders.

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